

# Experiment 'p\_4\_3\_e\_positions\_of\_vowels\_seq' Results

June 3, 2026

**Experiment outcome:** SUCCESS

**Bad responses:** 0

**Responses containing assume :** 0

**Responses containing expect :** 0

**Resolution attempts:** 1

**Hard fails (resolution):** 0

**Soft fails (resolution):** 0

**Verification attempts:** 1

## Problem Specification

**Problem name:** p\_4\_3\_e\_positions\_of\_vowels\_seq

**Natural language statement:** Write a method that takes a string and returns the positions of all vowels in the string.

**Method signature:** p\_4\_3\_e\_positions\_of\_vowels\_seq(s: string) returns (positions: seq<int>)

## Ensures

- forall p :: p in positions ==> 0 <= p < |s| && s[p] in ['a','e','i','o','u','A','E','I','O','U']
- forall i :: 0 <= i < |s| && s[i] in ['a','e','i','o','u','A','E','I','O','U'] ==> i in positions

## GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

### Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes a string and returns the positions of all vowels in the string.

The signature should be:

`p_4_3_e_positions_of_vowels_seq(s: string) returns (positions: seq<int>)`

The method should respect the following contract:

ensures forall p :: p in positions ==> 0 <= p < |s| && s[p] in ['a','e','i','o','u','A','E','I','O','U'], ensures forall i :: 0 <= i < |s| && s[i] in ['a','e','i','o','u','A','E','I','O','U'] ==> i in positions

Produce and show only the Dafny body of this method, including the curly braces that surround it. Do not show the signature nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

### GenAI → Program

**System fingerprint:** null

**ID:** chatcmpl-DmfTzspI62mS2NvV6pYE4i3AVDGGU

**Model:** gpt-5.5-2026-04-23

**Created at:** 1780491575

**Finish reason:** stop

**Usage:** 430 tokens in, and 1743 tokens out

```
{
  positions := [];
  var i := 0;
  while i < |s|
    invariant 0 <= i <= |s|
    invariant forall p :: p in positions ==> 0 <= p < i && s[p] in ['a','e','i','o','u','A','E','I','O','U']
    invariant forall j :: 0 <= j < i && s[j] in ['a','e','i','o','u','A','E','I','O','U'] ==> j in positions
  {
    if s[i] in ['a','e','i','o','u','A','E','I','O','U'] {
      positions := positions + [i];
    }
    i := i + 1;
  }
}
```

## Final Program

```
method p_4_3_e_positions_of_vowels_seq(s: string) returns (positions: seq<int>)
  ensures forall p :: p in positions ==> 0 <= p < |s| && s[p] in ['a','e','i','o','u'
    ↪ ', 'A','E','I','O','U']
  ensures forall i :: 0 <= i < |s| && s[i] in ['a','e','i','o','u','A','E','I','O',''
    ↪ 'U'] ==> i in positions
{
  positions := [];
  var i := 0;
  while i < |s|
    invariant 0 <= i <= |s|
    invariant forall p :: p in positions ==> 0 <= p < i && s[p] in ['a','e','i','o','u','A'
      ↪ ', 'E','I','O','U']
    invariant forall j :: 0 <= j < i && s[j] in ['a','e','i','o','u','A','E','I','O','U']
      ↪ ==> j in positions
  {
    if s[i] in ['a','e','i','o','u','A','E','I','O','U'] {
      positions := positions + [i];
    }
    i := i + 1;
  }
}
```

## Total Token Usage

Input tokens: 430

Output tokens: 1743

Reasoning tokens: 1536

Sum of ‘total tokens’: 2173

## Experiment Timings

**Overall Experiment** started at 1780491575261, ended at 1780491622142, lasting 46881ms (46.88 seconds)

**Iteration #1** started at 1780491575262, ended at 1780491622142, lasting 46880ms (46.88 seconds)